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Week 7 Cost Benefit Memo
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Cost-Benefit Memo: Model Selection for Phase 3

Verdict

I recommend carrying Logistic Regression into Phase 3 because it achieved the highest macro-F1, trained and predicted fastest, and was easier to understand than the more complex models.

Method

The analysis used 55,121 emergency department encounters and reused the same Week 6 train-test split so every model was evaluated on the same patients. The final feature set contained 209 triage-time features: the original eligible features plus age. Administrative fields and post-triage outcomes were excluded. Logistic Regression, untuned and tuned Random Forest, Gradient Boosting, a small neural network, and a Logistic Regression-Random Forest stacked model were trained on the same feature set. Models were compared using accuracy, macro precision, macro recall, macro-F1, ESI 1-3 recall, training time, prediction time, and ease of explanation. Macro-F1 measures performance across all five ESI urgency levels while giving each level equal importance, and recall measures the proportion of patients in each level that the model identified correctly.

Benchmark Results

Model	Accuracy	Macro Precision	Macro Recall	Macro-F1	Train (seconds)	Predict (ms)	Clarity
Logistic Regression	0.683	0.607	0.476	0.508	6.2	0.00289	High
Tuned Random Forest	0.618	0.459	0.523	0.481	241.7	0.09895	Moderate
Small MLP	0.687	0.623	0.452	0.478	21.1	0.00838	Low
Stacked Ensemble	0.605	0.438	0.627	0.448	720.2	0.10721	Low
Gradient Boosting	0.585	0.422	0.567	0.440	13.6	0.03145	Low-moderate
Untuned Random Forest	0.661	0.488	0.391	0.414	60.0	0.13445	Moderate

Table interpretation: Logistic Regression led macro-F1 at 0.508. The tuned Random Forest was second at 0.481 but took almost 39 times longer to train and over 34 times longer to make each prediction. The neural network had slightly higher accuracy, 0.687, but lower macro-F1, weaker ESI 1 recall, and low interpretability.

Cost-Benefit Analysis

1. Overall performance
Logistic Regression achieved the highest macro-F1 score at 0.508. This matters because macro-F1 gives each ESI level equal weight instead of allowing the large ESI 2 and ESI 3 groups to dominate the result. Its accuracy was 0.683, only 0.004 below the neural network, while its macro-F1 was 0.030 higher. The neural network therefore gained almost no accuracy but lost performance across the five ESI groups.

2. Critical-patient performance
Logistic Regression correctly identified 25.0% of ESI 1 patients and 62.6% of ESI 2 patients. Among the 16 ESI 1 patients in the test set, it classified 4 correctly, assigned 11 as ESI 2, and assigned 1 as ESI 4. This produced a 75.0% undertriage rate for ESI 1. Among 3,585 ESI 2 patients, it classified 2,246 correctly and undertriaged 1,336, producing a 37.3% undertriage rate.
The stacked model raised ESI 1 recall to 0.750, but this came with ESI 1 precision of 0.02 and macro-F1 of 0.448. In practical terms, it labeled many patients as ESI 1 to catch 12 of the 16 true ESI 1 cases. That creates a large false-alarm burden and weakens performance across the full ESI scale. Gradient Boosting reached the highest ESI 2 recall at 0.685, but its macro-F1 was only 0.440 and its ESI 3 recall fell to 0.486. The tuned Random Forest modestly improved ESI 1 recall to 0.312 and ESI 2 recall to 0.635, but its overall macro-F1 remained below Logistic Regression at 0.481.

3. Training and prediction cost
Logistic Regression trained in 6.2 seconds and made each prediction in 0.00289 milliseconds. The tuned Random Forest required 241.7 seconds to train and 0.09895 milliseconds per prediction. The stacked model required 720.2 seconds to train and 0.10721 milliseconds per prediction. These differences are small for one patient, but they become important when models are retrained, tested repeatedly, or used across a large hospital system.
The wider Random Forest search also showed a clear cost with no return. Increasing the search from 8 to 20 parameter combinations left the best cross-validation macro-F1 unchanged at 0.4681 and took 1,660.7 seconds, or 27.7 minutes. More search did not produce a better model.

4. Feature decisions
The feature experiment supported using the original eligible features plus age. This version gave Logistic Regression a macro-F1 of 0.508 and ESI 1 recall of 0.250. Engineered features plus age increased accuracy from 0.683 to 0.687, but macro-F1 fell to 0.491 and ESI 1 recall fell to 0.188. The added estimated mean arterial pressure feature also reduced Random Forest macro-F1 from 0.4138 to 0.3933. These results show that adding more derived variables did not improve the decision quality in this notebook.

Arguments Supporting Logistic Regression

1. It achieved the highest macro-F1 score at 0.508, which was 0.027 above the tuned Random Forest and 0.030 above the neural network.

2. It trained in 6.2 seconds and had the fastest prediction time at 0.00289 milliseconds per patient.
3. Its predictions can be traced to model coefficients, making the model easier to explain, audit, and update.
4. It provided strong ESI 2 and ESI 3 recall without the large false-alarm pattern produced by the stacked model.
5. Feature engineering and wider Random Forest tuning added cost without improving the selected performance measure.

Arguments Against Logistic Regression

1. It identified only 4 of 16 ESI 1 patients, giving ESI 1 recall of 0.250.
2. It undertriaged 75.0% of ESI 1 patients and 37.3% of ESI 2 patients in the test set.
3. It can miss feature combinations that tree-based and neural-network models can learn.
4. Its ESI 5 recall was 0.119, showing weak performance for the least urgent class.

Risks and Required Checks

The main risk is undertriage. Logistic Regression should not be used for automatic clinical decisions in its current form. Phase 3 should focus on reviewing every ESI 1 error, measuring how often ESI 1 and ESI 2 patients are assigned to less urgent levels, and testing whether class weighting or decision thresholds can improve critical-patient recall without creating excessive false alarms.

The ESI 1 result is based on only 16 test patients. Each patient changes ESI 1 recall by 6.25 percentage points, so this class must be tested on more cases. The model should also be tested on new train-test splits, later time periods, different patient groups, and data from another hospital before any deployment decision.

Fairness checks should compare errors across age, sex, race, ethnicity, language, and insurance groups. Missing-data checks should confirm that the model does not fail when common triage fields are absent. A clinical review should define which errors require a warning, a second review, or a rule-based safety check.

Final Recommendation

Use Logistic Regression for Phase 3. Keep tuned Random Forest as backup. Drop the stacked model. It flagged too many patients as critical to be useful.

Before deploying anything: test on more ESI 1 patients (16 is too few to trust), try to catch more ESI 1 and ESI 2 cases without hurting overall performance, and check the model for bias across patient groups. Logistic Regression is the best option so far, but it's not ready for real patients until this work is done and a clinician signs off.